

Hyuk Jun Kweon

CONTACT INFORMATION	27-218, Building 27, Seoul National University 1 Gwanak-ro, Gwanak-gu, Seoul 08826, South Korea	+82-10-9982-8055 kweon7182@snu.ac.kr
EMPLOYMENT	Seoul National University , Seoul, South Korea	
	BK Research Fellow	Jun 2025–Present
	University of Georgia , Athens, GA, USA	
	Limited Term Assistant Professor	Jan 2023–May 2025
	Massachusetts Institute of Technology , Cambridge, MA, USA	
	Research Specialist Limited (One-year bridge postdoc)	Aug 2021–Jul 2022
	National Institute for Mathematical Sciences , Daejeon, South Korea	
	Technical Research Personnel	Mar 2015–Apr 2018
	• Alternative military service • Managing a SageMath notebook server • Providing mathematical outreach programs	
EDUCATION	Massachusetts Institute of Technology , Cambridge, MA, USA	
	Ph.D. in Mathematics	Sep 2018–Jun 2021
	(Prior enrollment before compulsory military service)	Sep 2012–Aug 2014
	Advisor: Bjorn Poonen	
	Pohang University of Science and Technology , Pohang, South Korea	
	B.S. in Mathematics	Mar 2008–Feb 2012
	B.S. in Computer Science and Engineering	
	• Double Major • Summa Cum Laude	
PUBLICATIONS	A Uniform Construction of Cohomology Theories of Varieties via Fundamental Groupoids, <i>preprint</i>	
	Computing Picard Schemes (with Madhavan Venkatesh), <i>submitted</i>	
	Bounds on Torsion in Cohomology and an Effective Version of Deligne’s Gcd Theorem (with Madhavan Venkatesh), <i>to appear in Journal de Théorie des Nombres de Bordeaux</i>	
	Maximum Overlap Area of Several Convex Polygons Under Translation (with Honglin Zhu), <i>The 36th Canadian Conference on Computational Geometry (CCCG 2024)</i>	
	Maximum Overlap Area of a Convex Polyhedron and a Convex Polygon under Translation (with Honglin Zhu), <i>The 39th International Symposium on Computational Geometry (SoCG 2023)</i>	
	Bounds on the Torsion Subgroup Schemes of Néron-Severi Group Schemes, <i>Advances in Mathematics</i> 409 (2022): 108687.	
	Bounds on the Torsion Subgroups of Néron-Severi Groups, <i>Trans. Amer. Math. Soc.</i> 374 (2021), pp. 351–365.	
	Overlap of Convex Polytopes Under Rigid Motion (with Hee-Kap Ahn, Siu-Wing Cheng, and Juyoung Yon), <i>Computational Geometry</i> 47, no. 1 (2014): 15–24	

AWARDS AND HONORS	AMS-Simons Travel Grants	Jul 2023–Jun 2025
	Scholarship for Doctoral Study, Samsung Foundation	Sep 2012–Jun 2021
	Scholarship, STX Foundation	Mar 2011–Feb 2012
	Governmental Scholarship, Korea Student Aid Foundation	Mar 2008–Feb 2011
	6th Place, ACM-ICPC Korea Regional Programming Contest	Nov 2010
	4th Place, ACM-ICPC Korea Regional Programming Contest	Nov 2009
	5th Place, ACM-ICPC Korea Regional Programming Contest	Nov 2008
TEACHING	Calculus III for Engineering (MATH2500) Lecturer, University of Georgia	Jan 2025–May 2025
	Complex Variables (MATH4150) Lecturer, University of Georgia	Aug 2024–Dec 2024
	Precalculus (MATH1113) Lecturer, University of Georgia	Aug 2024–Dec 2024
	Calculus II for Science and Engineering (MATH2260) Lecturer, University of Georgia	Jan 2024–Apr 2024
	Calculus I for Science and Engineering (MATH2250) Lecturer, University of Georgia	Aug 2022–Dec 2023
	Precalculus (MATH1113) Lecturer, University of Georgia	Jan 2022–May 2022
	Multivariable Calculus (18.02) Recitation Leader, Massachusetts Institute of Technology	Feb 2021–May 2021
	Calculus (18.01A/18.02A) Recitation Leader, Massachusetts Institute of Technology	Sep 2020–Dec 2020
	Calculus of Several Variables (18.022) Recitation Leader, Massachusetts Institute of Technology	Sep 2019–Dec 2019
	Differential Equations (MATH200) Recitation Leader, Pohang University of Science and Technology	Mar 2012–Jun 2012
	Complex Variables (MATH310) Recitation Leader, Pohang University of Science and Technology	Aug 2011–Dec 2011
RESEARCH TALKS	International Conference for the 80th Anniversary of the Korean Mathematical Society	Jun 2026
	Seminar, Soongsil University	Mar 2026
	KIAS HCMC Algebra Seminar, Korea Institute for Advanced Study	Nov 2025
	POSTECH-PMI Number Theory Seminar, POSTECH	Nov 2025
	Rookies Workshop 2025, Seoul National University	Nov 2025
	QSMS Summer Workshop 2025, Goseong	Aug 2025
	Rookies Pitch, Seoul National University	Jun 2025
	Number Theory Seminar, University of Georgia	May 2025
	Canadian Conference on Computational Geometry, Brock University	Jul 2024
	Palmetto Number Theory Series XXXVI, Clemson University	Oct 2023
	Number Theory Seminar, University of Georgia	Apr 2023

	Southern Regional Number Theory Conference, Louisiana State University	Mar 2023
	Number Theory Seminar, University of Georgia	Aug 2022
	Korea Institute for Advanced Study	Oct 2022
	POSTECH-PMI Number Theory Seminar, POSTECH	Apr 2022
	Johns Hopkins Junior Number Theory Days, Johns Hopkins University	Dec 2021
	2021 Korean Mathematical Society Annual Meeting	Oct 2021
	Ohio State Number Theory Seminar, Ohio State University	Mar 2021
	AMS Special Session, Spring Eastern Sectional Meeting, Brown University	Mar 2021
	Binghamton Arithmetic Seminar, Binghamton University	Mar 2021
	Korea Institute for Advanced Study	Jan 2020
EXPOSITORY TALKS	Proof of the amplification principle 2	May 2022
	STAGE, Massachusetts Institute of Technology	
	Alterations	Nov 2020
	STAGE, Massachusetts Institute of Technology	
	Formal Groups as Functors	Sep 2019
	STAGE, Massachusetts Institute of Technology	
	Division by 2 on Hyperelliptic Curves	Dec 2018
	Preprint Seminar, Massachusetts Institute of Technology	
	Review of Étale Cohomology	Sep 2018
	STAGE, Massachusetts Institute of Technology	
	On the Fary-Milnor Theorem, following the existing literature	Nov 2010
ACADEMIC SERVICE	SCV Seminar, Pohang University of Science and Technology	
	Co-organizer of STAGE	Spring 2022
	(Seminar on Topics in Arithmetic, Geometry, Etc.), MIT	
	Co-organizer of STAGE	Fall 2021
	(Seminar on Topics in Arithmetic, Geometry, Etc.), MIT	
	Co-organizer of STAGE	Fall 2020
	(Seminar on Topics in Arithmetic, Geometry, Etc.), MIT	